

Zilei Shao

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EDUCATION

University of California, Los Angeles

Ph.D. in Computer Science | Advisor: [Guy Van den Broeck](#)

Los Angeles, CA

Sep. 2024 - Present

Harvey Mudd College

B.S. in Joint Major of Computer Science and Mathematics | GPA: 3.94/4.0

Claremont, CA

Aug. 2020 - May 2024

RESEARCH EXPERIENCE

StarAI Lab, Los Angeles, UCLA

Graduate Researcher Advised by Professor Guy Van den Broeck

May 2024 – Present

Los Angeles, CA

- Developing techniques to improve training efficiency and performance of deep generative models by integrating **tractable probabilistic models** into modern neural architectures.
- Exploring tokenization in Large Language Models (LLMs) as a **neurosymbolic** problem.

Uncertainty in Reinforcement Learning, L.A.C.E. Lab, Harvey Mudd College

Undergraduate Researcher Advised by Professor Erin Talvitie

Jun. 2022 – Jul. 2023

Claremont, CA

- Developed bounding-box inference to propagate uncertainty in environment, increasing the performance of a model-based reinforcement learning algorithm Model-based Value Expansion (MVE) by over 100%.

PUBLICATIONS

- [1] Anji Liu, [Zilei Shao](#), Guy Van den Broeck. Rethinking Probabilistic Circuit Parameter Learning. *Under Review*. [[PDF](#)]
- [2] [Zilei Shao](#), Anji Liu, Guy Van den Broeck. Zero-Variance Gradients for Variational Autoencoders. *In NeurIPS 2025 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2025. [[PDF](#)]
- [3] Renato L. Geh*, [Zilei Shao](#)*, Guy Van den Broeck. Adversarial Tokenization. *In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. (**ACL 2025 Main Conference**) [[Webpage](#), [PDF](#)]
- [4] Erin J. Talvitie, [Zilei Shao](#), Huiying Li, Jinghan Hu, Jacob Boerma, Rory Zhao, Xintong Wang. Bounding-Box Inference for Error-Aware Model-Based Reinforcement Learning. *Reinforcement Learning Journal*, vol. 1, no. 1, 2024. (**RLC 2024**) [[PDF](#)]

PROFESSIONAL EXPERIENCE

Waymo LLC

Jun. 2025 – Sep. 2025

- Software Engineer Intern in Predictive Planner team.
- Improved motion prediction models by developing dynamic model capacities for more accurate agent forecasting.

TALKS

- [1] Adversarial Tokenization, Zilei Shao. *Shang Data Lab at UCSD, April 2025*.
- [2] Range-to-Range Regression: Propagating Uncertainty in Environment Model, Zilei Shao, Erin Talvitie (advisor). *Harvey Mudd College Summer CS Talk, 2023*.

TEACHING EXPERIENCES

- CS161 Fundamentals of Artificial Intelligence, UCLA, 2025 Fall.
- CS5 Introduction to Computer Science, Harvey Mudd College, 2024 Spring, 2023 Fall.
- MATH179 Mathematics of Big Data, Harvey Mudd College, 2024 Spring.
- CS81 Computability and Logic, Harvey Mudd College, 2023 Spring.
- CS70 Data Structures and Program Development, Harvey Mudd College, 2022 Fall.
- CS181Y CS for Inquiry, Harvey Mudd College, 2022 Spring.
- CS60 Principles of Computer Science, Harvey Mudd College, 2021 Fall.